

Voice-Controlled Scribe Bot for Assisting the Disabled

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I. Introduction

The ability to write is a fundamental aspect of communication, education, and daily functioning. However, for individuals with physical disabilities affecting motor control, writing can be a significant challenge. This project presents the development of a voice-controlled scribe bot, a system designed to convert spoken language into written text on paper using a mechanical plotting mechanism. The primary objective of this system is to provide an assistive solution that enables users to write simply by speaking. What makes this project unique is that it bridges the gap between speech recognition and physical actuation. While speech-to-text software is widely available, there are virtually no accessible systems that directly translate voice input into real-time handwritten output using stepper motors. A major challenge encountered during development was the absence of any existing software or framework capable of controlling stepper motors through voice commands while simultaneously handling text formatting on a physical medium such as an A4 sheet. Therefore, the entire system—including voice processing, layout control, and motor actuation—was developed from scratch.

II. Literature Survey

Assistive technologies for individuals with motor impairments have seen significant advancements in recent years, particularly in the domains of speech recognition and human-computer interaction. One of the most widely adopted solutions is speech-to-text conversion, which enables users to generate digital text through voice commands. Systems such as Google Speech-to-Text and CMU Sphinx have demonstrated high accuracy in transcribing spoken language into editable text formats. However, these systems are primarily limited to digital environments and do not address the need for physical writing.

Research in the field of handwriting automation has explored the use of robotic plotters and CNC-based systems for reproducing text and drawings. Devices such as pen plotters and modified CNC machines utilize stepper motors and control algorithms to generate precise movements on paper. While these systems are effective in replicating pre-defined inputs (e.g., G-code instructions), they lack real-time adaptability and are not designed for direct interaction through natural interfaces such as voice.

In parallel, voice-controlled robotic systems have been developed for applications such as home automation, robotic arms, and assistive mobility devices. These systems typically integrate speech recognition modules with microcontrollers like Arduino to trigger predefined actions. However,

most implementations are limited to discrete command execution (e.g., “move forward”, “turn left”) rather than continuous tasks such as writing structured text.

Some studies have attempted to bridge the gap between speech and physical output by combining speech recognition with robotic manipulators. However, these approaches often focus on simple drawing tasks or symbolic representations rather than structured handwriting. Moreover, they rely heavily on pre-trained datasets and predefined motion paths, limiting their flexibility and scalability.

A key limitation across existing works is the lack of an integrated system that converts continuous speech input into dynamically formatted, real-time handwritten output on a physical medium. Specifically, there is minimal research addressing:

- Real-time mapping of spoken language to spatial writing trajectories,
- Layout management for writing on constrained surfaces such as A4 sheets,
- Synchronization between speech processing and multi-axis motor control.

Therefore, this project aims to address these gaps by developing a fully integrated voice-controlled scribe system. Unlike existing solutions, the proposed system combines speech recognition, text processing, and stepper motor-based actuation into a unified pipeline capable of generating handwritten text directly from voice input. This makes it particularly suitable as an assistive tool for individuals with motor disabilities, enabling them to perform writing tasks independently.

III. Proposed Method

A. Electrical System

The electrical subsystem forms the control backbone of the scribe bot. It is built around an Arduino microcontroller interfaced with a CNC shield, which simplifies the process of controlling multiple stepper motors. The A4988 stepper motor drivers are mounted on the CNC shield and are responsible for driving the X and Y axis motors with precise step control. The system operates by receiving G-code instructions from a computer via serial communication. These commands are interpreted by the Arduino (running GRBL firmware), which then generates appropriate pulse signals for the stepper motors. The servo motor, used for controlling the pen’s vertical movement, is driven using PWM signals. A 12V power supply ensures that sufficient current is available for motor operation while maintaining stable performance.

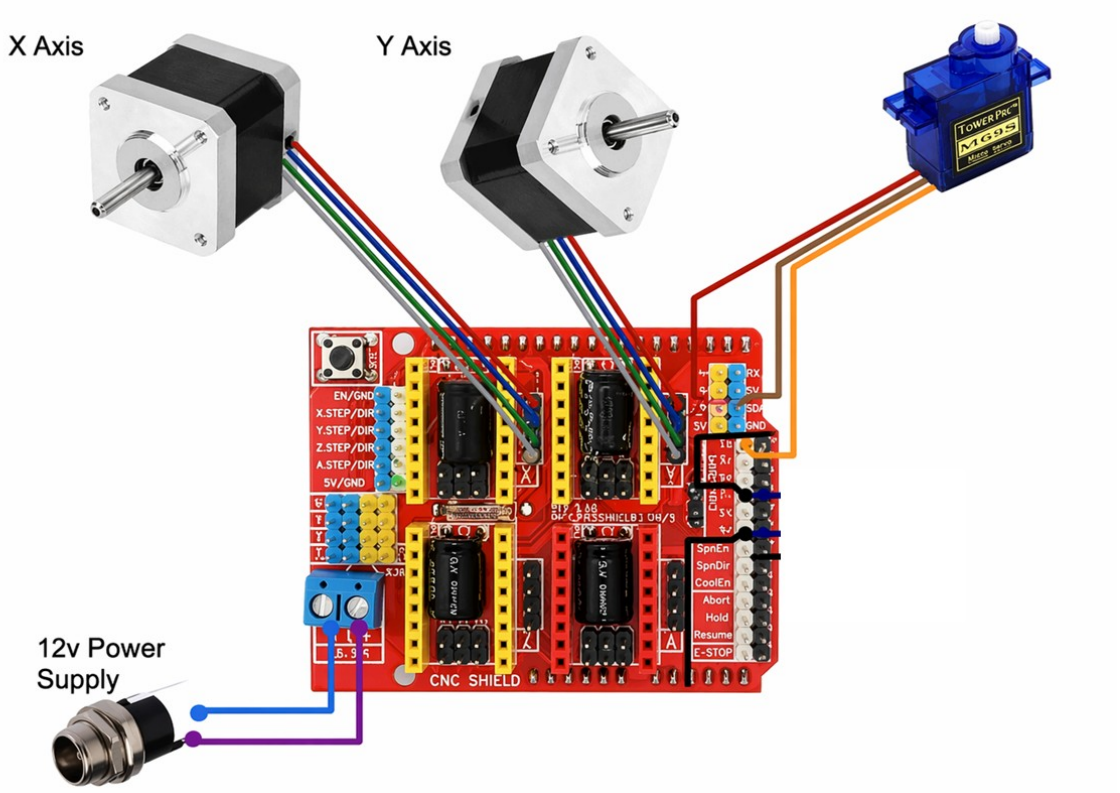


Fig. 1. Comparison between ground-truth diffusion map and predicted diffusion map in simulated data

B. Mechanical Design

The mechanical structure of the scribe bot is designed to achieve controlled motion in a two-dimensional plane. The base and supporting structures were fabricated using acrylic sheets, which were precisely cut using a laser cutting machine to match the dimensions of the motors and other components. These acrylic parts serve as the framework for mounting and alignment. Two perpendicular axes (X and Y) were constructed using steel rods as guides, ensuring smooth and stable motion. The X-axis serves as the base movement, while the Y-axis is mounted on top of it and carries the pen assembly. A screw rod was incorporated into the X-axis to enhance structural stability and reduce vibrations during operation. Motion transmission is achieved using timing belts and timing pulleys, which provide smooth and backlash-free movement. Stepper motors with a step angle of 3.6° were chosen due to their high precision and suitability for incremental positioning tasks such as writing. The pen lifting mechanism is implemented using a servo motor, which provides controlled angular motion within a limited range. The pen holder was

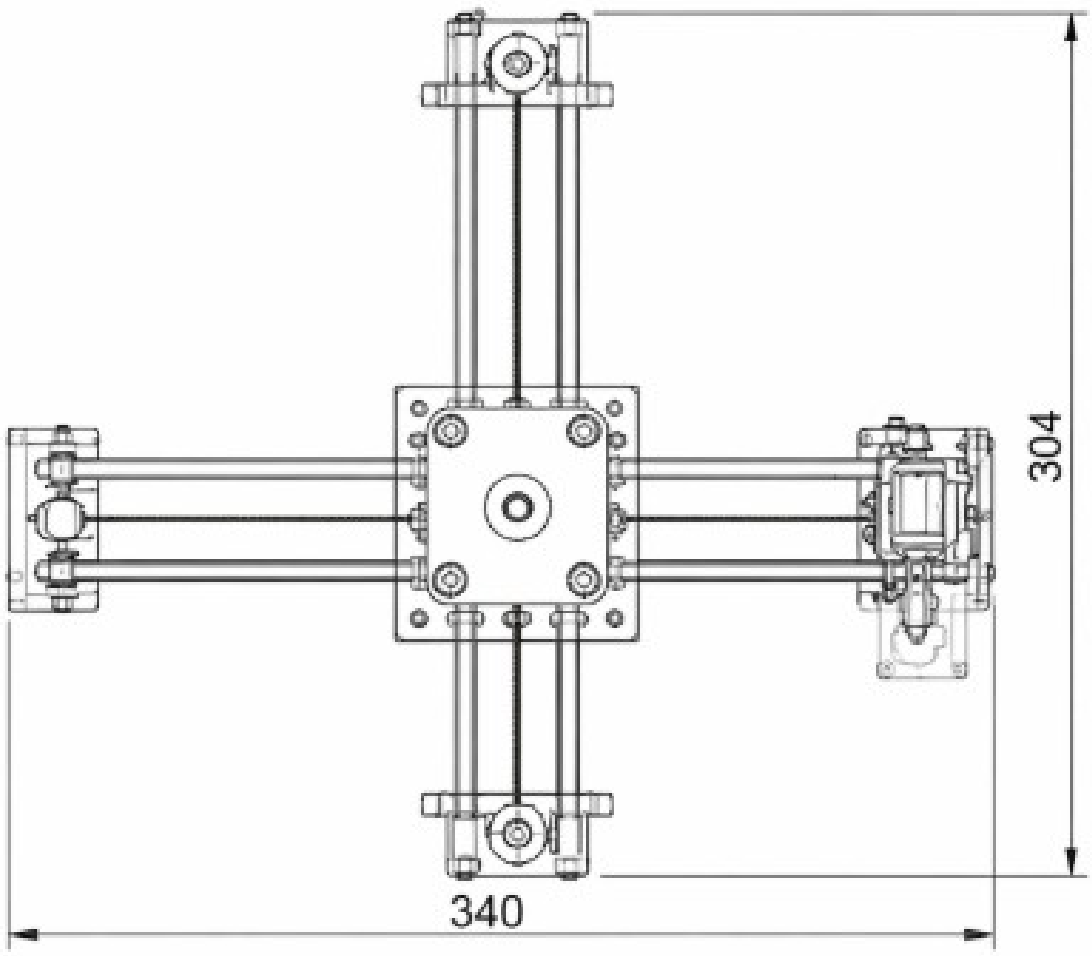


Fig. 2. Comparison between ground-truth diffusion map and predicted diffusion map across different experimental data.

constructed using pen refills and supported with springs to maintain consistent pressure between the pen and the paper. This ensures continuous contact during writing. A brush pen was selected instead of a conventional pen due to its flexible nib, which improves writing consistency even with minor surface irregularities.

C. Software and Control Logic

The software component is the most critical part of the system, as it integrates voice recognition with motor control. A custom Python program was developed to handle the entire pipeline, including capturing voice input, converting it to text, generating motion commands, and sending them to the Arduino. The program uses a microphone to capture audio input and processes it using a speech recognition library. The recognized text is then passed through a custom algorithm

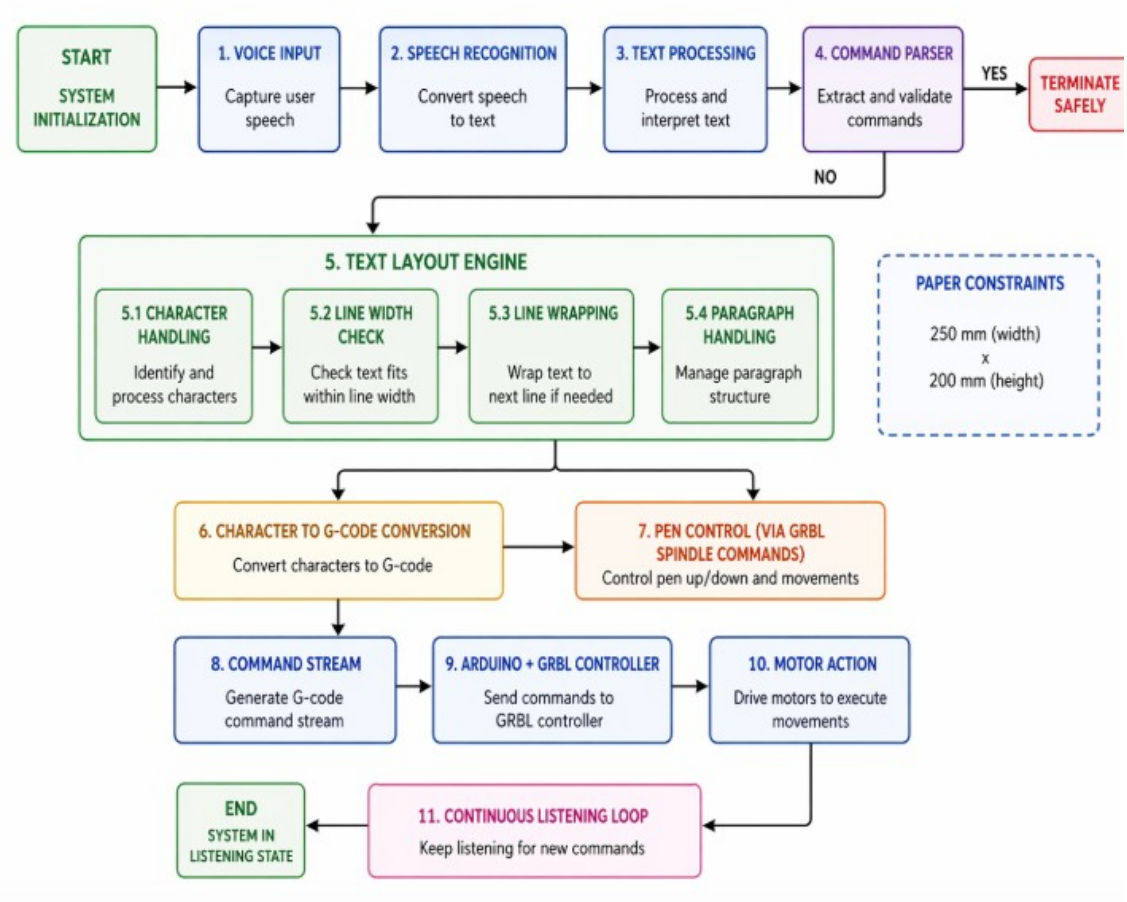


Fig. 3. Comparison between ground-truth diffusion map and predicted diffusion map across different experimental data.

that converts each character into a series of G-code instructions. These instructions define the exact path that the pen must follow to draw each letter. A key innovation in this system is the implementation of a text layout engine, which ensures that the written output fits within the physical boundaries of the paper. The software automatically handles character spacing, word spacing, line wrapping, and paragraph transitions. For example, when the current line exceeds the maximum width, the system performs a carriage return and continues writing on the next line. Additionally, voice commands such as “paragraph” allow the user to insert spacing between lines. The pen control mechanism is implemented using GRBL spindle commands, where different PWM values correspond to pen-up and pen-down positions. The system continuously listens for voice input, processes it in real time, and sends the corresponding commands to the Arduino. Special commands such as “stop” terminate the writing process safely.

IV. Results

The system was successfully assembled by integrating the electrical components with the mechanically designed structure, resulting in a fully functional voice-controlled scribe bot. After completing the fabrication and connections, the device was tested under different conditions to evaluate its performance. Writing tests were carried out on paper sheets of varying dimensions, with the largest test surface being 250 mm × 200 mm, to verify the effectiveness of the layout algorithm and motion control. The system was able to accurately interpret voice input and convert it into written output while maintaining proper alignment and spacing across the writing surface. However, one limitation observed during testing was the presence of vibrations generated by the stepper motors during operation. These vibrations slightly affect the smoothness of motion, leading to minor distortions in the written output and a small reduction in overall writing quality.

V. Application

This system has significant potential in assistive technology, particularly for individuals with limited motor abilities. It can also be used in educational environments to demonstrate the integration of mechanical systems with artificial intelligence. Additionally, it serves as a low-cost alternative to plotting machines and can be extended for artistic or design applications.

VI. Conclusion

The voice-controlled scribe bot successfully demonstrates the integration of speech recognition, embedded systems, and mechanical design to create a functional assistive device. By enabling users to write using only their voice, the system provides an innovative solution to a real-world problem. Future improvements could include enhanced handwriting styles, offline speech recognition capabilities, and support for multiple languages, further increasing the usability and accessibility of the system.